

Traffic analysis dataset

Project description

1. Project Description

This project analyzes traffic accident data to understand road safety, accident distribution, contributing factors, and high-risk bottlenecks.

Using the **Traffic Accidents dataset**, we will clean and process the data, perform exploratory analysis, and create an interactive dashboard to generate insights that can help transportation authorities, urban planners, and emergency services improve road safety, reduce accident severity, and optimize traffic management and resource allocation.

The project will focus on analyzing traffic events such as accident volume, severity levels, regional or hotspot distribution, environmental conditions (like weather and time of day), and long-term safety trends.

2. Group Members & Roles

Member	Role
Ahmed Hossam Mohamed Nabil (Team leader)	1. Medallion architecture (Bronze, Silver, Gold) 2. Power-bi deployment (landing page, overview page) 3. Documentation 4. EDA
Youssef Amr	1. Medallion architecture (Silver) 2. Power-bi deployment (Crash cause page, dark mode) 3. EDA 4. initial dashboard design using figma
Eslam Jabr	1. Medallion architecture (Bronze, Silver) 2. Machine learning model 3. EDA 4. Power-bi deployment (Ai insights page, Severity page)

Fagr Abdullah	1. Power-bi deployment (Road analysis page) 2. power bi overall dashboard design and color palette
Basmala sabri	1. Presentation 2. Power-bi deployment (Weather page)
Salma adel	

3. Project Objectives

The main objectives of this project are:

1. **Analyze** traffic accident patterns and collision frequency.
 2. **Identify** contributing factors to accident severity and road hazards (e.g., weather, lighting, road conditions).
 3. **Evaluate** high-risk times and seasonal trends in traffic incidents.
 4. **Identify** patterns in the types of vehicles involved, collision types, and victim demographics.
 5. **Provide** data-driven recommendations to improve road safety, traffic management, and emergency response infrastructure.
 6. **Build** a dashboard that clearly communicates insights to stakeholders and urban planners.
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4. Tools & Technologies

The following tools will be used during the project:

Tool	Purpose
Power point	For designing the presentation.
Python (Pandas, NumPy)	1. Used in medallion architecture (Bronze, Silver, Gold) to clean the data and create a star schema. 2. Used for Exploratory data analysis (EDA).
Figma	Initial dashboard design and color palette selection.

Power BI	Creating DAX measures and visualizations.
GitHub	Project collaboration and version control
Jupyter Notebook	Analysis documentation

5. Milestones & Deadlines

Milestone	Description	Deadline
Dataset Understanding	Explore dataset structure and variables	Week 1
Medallion architecture using python	Read raw data, perform data cleaning, creating a star schema data model	Week 2
Exploratory Data Analysis (EDA) using python	Identify trends and patterns	Week 3
Initial dashboard design using figma	Choosing color palette, creating visuals, designing the overall pages initial design	Week 4
Dashboard Development using power bi	Creating DAX measures and visualizations	Week 5
Creating documentation and presentation	Deliver documentation and presentation	Week 6

6. Business Questions

1. What is the total number of traffic accidents recorded?
2. What are the peak times of day or days of the week for traffic incidents?
3. Which environmental conditions (e.g., weather, lighting, road surface) are most frequently associated with accidents?
4. What is the distribution of accident severity (e.g., minor, severe, fatal)?
5. Which types of collisions or maneuvers (e.g., rear-end, sideswipe, turning) are most common?
6. What trends can be observed in accident frequency over different months or seasons?

7. Key Performance Indicators (KPIs)

A. Data Cleaning & Processing KPIs

KPI	Description	Target
Missing Data Handling Rate	Percentage of missing values handled	100%
Duplicate Records Removed	Duplicate rows identified and removed	100%
Data Consistency	Correct formatting for dates, IDs, and categorical fields	100%
Data Validation Accuracy	Clean dataset passes validation checks	≥ 95%

B. Analysis & Insights KPIs

KPI	Description	Target
Business Questions Answered	Percentage of questions successfully answered	≥ 90%
Insights Generated	Number of meaningful insights identified	≥ 5
Data-driven Recommendations	Recommendations supported by analysis	≥ 3

C. Traffic analysis KPIs

Casualty & Injury Breakdown

- Total Accidents
- Total Injuries
- Total_Fatal_injuries
- Incapacitating_Injuries
- Non_Incapacitating_Injuries
- Not_Visible_Injuries

Severity & Risk Analytics

- Severity rate %
- Severity_Index

Model & Data Performance

- Accuracy

D. Visualization & Reporting KPIs

KPI	Description	Target
Dashboard Load Time	Dashboard responsiveness	< 3 seconds
Dashboard Usability	Users navigate without assistance	≥ 80%
Number of Visualizations	Charts used to explain insights	≥ 8
Interactivity	Filters and slicers included	Yes

E. Final Documentation & Presentation KPIs

KPI	Description	Target
Final Report Completeness	All sections included	100%
Actionable Recommendations	Business recommendations generated	≥ 3
Presentation Quality	Clear storytelling and insights	≥ 80% feedback score

8. Dashboard Visualizations

The dashboard will include:

- **Accident & Injury Summary (KPI Cards):** High-level metrics displaying *Total Accidents*, *Total Injuries*, *Total Fatal Injuries*, and *Severity Rate %*.
- **Accidents Over Time (Line Chart):** A historical trend chart showing monthly or seasonal distributions to track whether accident rates are improving or worsening over time.
- **Accidents by weather and crash severity (Column Chart):** A vertical bar chart tracking accident frequency by the level of severity of the accident to easily identify how severe the accidents were.
- **Injury Severity Breakdown (Donut Chart):** A circular visual separating the proportion of incident outcomes (*Fatal*, *Incapacitating*, *Non-Incapacitating*, and *Not Visible*).
- **Environmental & Road Conditions (Stacked Bar Chart):** A breakdown showing how different weather conditions impact accident severity levels.

- **Accidents by road surface (Treemap):** A nested, space-efficient chart to display how the road surface impacts the number of accidents.
 - **Impact of Lighting (column chart):** A vertical chart comparing total accidents across various lighting situations (e.g., daylight vs. dark-unlit).
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9. Generated Insights

The project aims to uncover insights such as:

1. **Peak risk periods:** Identifying the specific hours of the day and days of the week when traffic accidents are most likely to occur.
 2. **Environmental danger catalysts:** Uncovering how adverse weather, poor lighting, and hazardous road surface conditions correlate with a higher accident frequency.
 3. **Severity drivers:** Isolating the critical factors (such as collision types or vehicle involvements) that transition a standard traffic incident into a fatal or incapacitating event.
 4. **Vehicle and collision patterns:** Determining which types of vehicles and maneuver habits are most frequently tied to high-impact accidents.
 5. **Safety trends over time:** Revealing macro-level patterns to show whether road safety conditions are stabilizing, deteriorating, or improving across different months and seasons.
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10. Business Recommendations

1. Target Resource Allocation During Peak Hours

- **Recommendation:** Deploy traffic enforcement and rapid-response emergency units to known critical corridors during the identified peak accident hours (e.g., morning and evening rush hours).
- **Impact:** Faster response times to incidents can significantly reduce the escalation of injuries and prevent secondary collisions caused by bottleneck congestion.

2. Implement Weather and Lighting Infrastructure Upgrades

- **Recommendation:** For environments where dark or unlit conditions drive up the *Severity Index*, invest in high-visibility street lighting, reflective road markings, and smart, dynamic warning signs that trigger during adverse weather.
- **Impact:** Proactively mitigates environmental hazards before they result in severe or incapacitating injuries.

3. Launch Target Safety Campaigns for High-Risk Collision Types

- **Recommendation:** Develop targeted public awareness campaigns and stricter traffic enforcement focusing on the most common collision types and vehicle categories revealed by the *Treemap* analysis (e.g., anti-tailgating campaigns if rear-end collisions dominate).
- **Impact:** Directly addresses poor driver habits and maneuvers that contribute heavily to overall accident volume.

4. Policy and Vehicle Regulations Review

- **Recommendation:** Analyze the correlation between specific vehicle types and high *Severity Rates %* to inform policy decisions. This could include updating safety inspection requirements for high-risk commercial fleets or adjusting speed limits for specific vehicle classes during high-risk seasons.
- **Impact:** Lowers the long-term trend of fatal and incapacitating injuries by addressing structural risks in vehicle use.

5. Establish a Continuous Monitoring Framework

- **Recommendation:** Utilize the interactive Power BI dashboard as a "Single Source of Truth" for monthly safety reviews. If the *Severity Index* or *Total Accidents* spike in a given month, trigger an immediate cross-departmental audit to investigate the cause.
- **Impact:** Shifts public safety efforts from a **reactive** posture to a **proactive**, continuous improvement model.